

WORK & ENERGY

Complete Concepts • Formulae • Examples • Practice Questions • Solutions

Chapter Snapshot

Understand when work is done, positive/negative/zero work, energy and its forms, kinetic and potential energy, law of conservation of energy, power, commercial unit of energy, and numerical problem-solving.

1. Work

In science, **work is done when a force causes displacement** in the direction of the force. Applying force without displacement does not count as mechanical work.

$$\text{Work} = \text{Force} \times \text{Displacement}$$

$$W = F \times s$$

SI unit of work = **joule (J)**. One joule of work is done when a force of 1 N produces a displacement of 1 m in its direction.

Types of Work

Positive work: force and displacement are in the same direction.

Negative work: force acts opposite to displacement.

Zero work: no displacement, or force is perpendicular to displacement.

Examples of Work

- Pushing a box and moving it forward: positive work.
- Friction slowing a moving object: negative work.
- Pushing a wall that does not move: zero work.
- Carrying a bag horizontally at constant height: work done by the upward force on the bag is zero because displacement is horizontal.

2. Energy

Energy is the capacity to do work. An object possessing energy can perform work.

$$\text{SI unit of energy} = \text{joule (J)}$$

Common forms include mechanical, heat, light, sound, electrical and chemical energy. In this chapter, the main mechanical forms are kinetic and potential energy.

3. Kinetic Energy

Kinetic energy is the energy possessed by an object due to its motion.

$$KE = \frac{1}{2}mv^2$$

m = mass in kg and v = speed in m/s. If speed doubles, kinetic energy becomes four times because KE is proportional to v^2 .

Key Idea

A fast-moving car has more kinetic energy than the same car moving slowly. A heavier object moving at the same speed also has greater kinetic energy.

4. Potential Energy

Potential energy is the energy possessed by an object because of its position or configuration. Gravitational potential energy near Earth's surface depends on height.

$$PE = mgh$$

m = mass (kg), g = acceleration due to gravity (m/s^2), h = height (m).

5. Law of Conservation of Energy

Law

Energy can neither be created nor destroyed. It can only be transformed from one form to another; therefore, the total energy of an isolated system remains constant.

Example: When a stone falls, its potential energy decreases while kinetic energy increases. Ignoring air resistance, total mechanical energy remains constant.

6. Power

Power tells us how fast work is done or how fast energy is transferred.

$$\text{Power} = \text{Work done} / \text{Time taken}$$

$$P = W/t$$

SI unit of power = **watt (W)**. 1 watt = 1 joule per second. Larger unit: 1 kilowatt (kW) = 1000 W.

7. Commercial Unit of Energy

Electrical energy used in homes is commonly measured in **kilowatt-hour (kWh)**, often called one **unit** of electricity.

$$1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$$

Meaning of 1 Unit

If a 1 kW appliance runs for 1 hour, it consumes 1 kWh = 1 unit of electrical energy.

Solved Examples

Example 1: Work

A force of 50 N moves a box by 4 m in the same direction. Find work.

$$W = Fs = 50 \times 4 = \mathbf{200 \text{ J}}.$$

Example 2: Zero work

A person pushes a rigid wall but it does not move. Displacement = 0, therefore $W = F \times 0 = \mathbf{0 \text{ J}}$.

Example 3: Kinetic energy

Find KE of a 2 kg object moving at 5 m/s.

$$KE = \frac{1}{2}mv^2 = \frac{1}{2} \times 2 \times 25 = \mathbf{25 \text{ J}}.$$

Example 4: Potential energy

A 3 kg object is lifted to 10 m. Take $g = 9.8 \text{ m/s}^2$.

$$PE = mgh = 3 \times 9.8 \times 10 = \mathbf{294 \text{ J}}.$$

Example 5: Power

A machine does 600 J work in 20 s. Find power.

$$P = W/t = 600/20 = \mathbf{30 \text{ W}}.$$

Example 6: Electrical energy

A 2 kW heater runs for 3 hours. Energy used = $2 \times 3 = \mathbf{6 \text{ kWh (6 units)}}$.

Practice Questions

1. Define work. State its SI unit.
2. When is work said to be positive, negative and zero?
3. A force of 100 N moves an object 5 m in its direction. Calculate work done.
4. Why is no mechanical work done when pushing a wall that does not move?
5. Define energy and name its SI unit.
6. Derive or state the formula for kinetic energy.
7. Find the kinetic energy of a 4 kg body moving at 3 m/s.
8. What is potential energy? Write its formula near Earth's surface.
9. Find the potential energy of a 5 kg object at a height of 8 m. Take $g = 10 \text{ m/s}^2$.
10. State the law of conservation of energy with one example.
11. Define power and write its SI unit.
12. A person does 1200 J work in 30 s. Find power.
13. What is 1 kWh? Convert it into joules.
14. A 100 W bulb glows for 10 hours. Find electrical energy consumed in kWh.
15. Explain energy transformation in a freely falling object.

Answers & Solutions

1. Work is done when a force produces displacement. SI unit: joule (J).
2. Positive: same direction; negative: opposite directions; zero: no displacement or perpendicular force-displacement.
3. $W = Fs = 100 \times 5 = \mathbf{500 \text{ J}}$.
4. Because displacement is zero, $W = F \times 0 = \mathbf{0 \text{ J}}$.
5. Energy is the capacity to do work. SI unit: joule (J).
6. Kinetic energy = $\frac{1}{2}mv^2$.
7. $KE = \frac{1}{2} \times 4 \times 3^2 = 2 \times 9 = \mathbf{18 \text{ J}}$.
8. Potential energy is energy due to position or configuration. Near Earth: $PE = mgh$.
9. $PE = mgh = 5 \times 10 \times 8 = \mathbf{400 \text{ J}}$.
10. Energy cannot be created or destroyed, only transformed. Example: in a falling object, PE changes into KE.
11. Power is work done per unit time. SI unit: watt (W).
12. $P = W/t = 1200/30 = \mathbf{40 \text{ W}}$.
13. 1 kWh is energy used by a 1 kW appliance in 1 hour. $\mathbf{1 \text{ kWh} = 3.6 \times 10^6 \text{ J}}$.

14. $100\text{ W} = 0.1\text{ kW}$. Energy = $0.1 \times 10 = \mathbf{1\text{ kWh} = 1\text{ unit}}$.

15. As an object falls, gravitational potential energy decreases and kinetic energy increases. Total mechanical energy remains constant if air resistance is ignored.

Quick Revision

Work	$W = Fs$ (when force and displacement are in same direction)
Unit of work & energy	joule (J)
Kinetic Energy	$KE = \frac{1}{2}mv^2$
Potential Energy	$PE = mgh$
Conservation of Energy	Energy changes form; total is conserved
Power	$P = W/t$
Unit of power	watt (W)
Commercial unit	$1\text{ kWh} = 3.6 \times 10^6\text{ J}$

Exam Tip: In numericals, always write Given → Formula → Substitution → Final answer with correct unit.